



JRS TUTORIALS

TEST PAPER' 2026–27

XII–IIT-JEE(Main)

PART TEST – VI

STAR

Time: 3 hours

M.M. : 300

INSTRUCTIONS :

A. General :

1. The Test Booklet consists of **75** questions. The maximum marks are **300**.
2. There are **Three** parts in the question paper A, B, C consisting of **Physics, Chemistry** and **Mathematics** having 25 questions in each part consists of **Two Sections** of equal weightage.
3. In **Section I** (Total Marks: 80), for each question you will be awarded **4 marks** if you darken **ONLY** the bubble corresponding to the correct answer and **zero marks** if no bubble is darkened. In all other cases, **minus one (–1) mark** will be awarded.
4. In **Section II** (Total marks: 20), for each question you will be awarded **4 marks** if you darken **ONLY** the bubble corresponding to the correct answer and **zero marks** if no bubble is darkened. In all other cases, **minus one (–1) mark** will be awarded.
5. **Marking Scheme** – For each correct answer you will be awarded **4(four) marks** and for each incorrect response **1 mark** will be deducted.
6. The Question Paper contains blank pages for your rough work. No additional sheets will be provided for rough work.
7. Blank papers, clip boards, log tables, slide rule, calculators, cellular phones, pagers and electronic gadgets, in any form, are **NOT** allowed.
8. Write your **Name** and **Roll No.** in the space provided at the bottom of this sheet.

B. Answering :

Section I –

9. Every question has **4** choices for its answer (1), (2), (3) and (4).
10. Darken the bubble, with black ball point pen, corresponding to what you consider to be the most appropriate answer, from among the four choices in your answer sheet.
11. **Do not fold or make any stray marks on the Answer Sheet.**

Section II –

This section contains **5 questions**. This answer to each of the questions is **integer type**, ranging from **0 to 999**. The bubble corresponding to the correct answer is to be darkened in the ORS.

Roll No. :

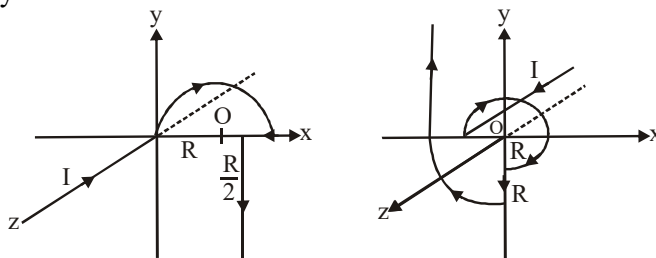
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PART – A (PHYSICS)
SECTION – I (Total Marks : 80)
(Single Correct Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct

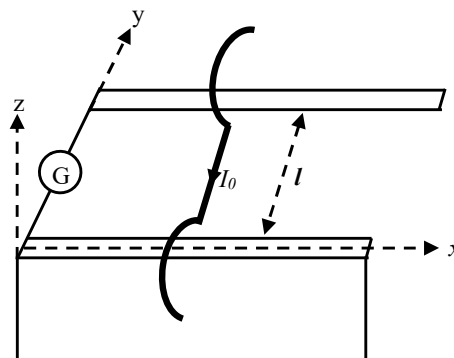
Take $g = 10 \text{ m/sec}^2$ unless otherwise specified.

1. The magnetic field at O in vector form for each of the following current configurations are respectively



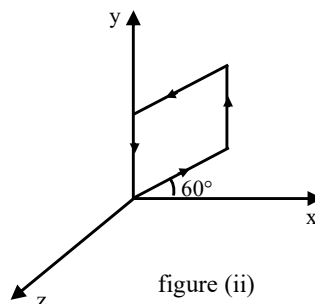
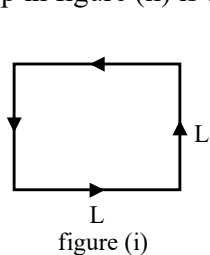
1. $\left(\frac{-\mu_0 I}{2\pi R}\right)(\hat{j}) + \left(\frac{\mu_0 I}{2R} + \frac{\mu_0 I}{2\pi R}\right)(-\hat{k})$ and $\frac{\mu_0 I}{4\pi R}(\hat{j}) + \frac{\mu_0 I}{8\pi R}(-\hat{k}) + \left(\frac{\mu_0 I}{16R} + \frac{3\mu_0 I}{16R}\right)(-\hat{k})$
 2. $\left(\frac{-\mu_0 I}{4\pi R}\right)(\hat{j}) + \left(\frac{\mu_0 I}{4R} + \frac{\mu_0 I}{2\pi R}\right)(-\hat{k})$ and $\frac{\mu_0 I}{2\pi R}(\hat{j}) + \frac{\mu_0 I}{4\pi R}(-\hat{k}) + \left(\frac{\mu_0 I}{8R} + \frac{3\mu_0 I}{8R}\right)(-\hat{k})$
 3. $\left(\frac{-\mu_0 I}{4\pi R}\right)(\hat{j}) + \left(\frac{\mu_0 I}{4R} + \frac{\mu_0 I}{2\pi R}\right)(-\hat{k})$ and $\frac{\mu_0 I}{4\pi R}(\hat{j}) + \frac{\mu_0 I}{8\pi R}(-\hat{k}) + \left(\frac{\mu_0 I}{16R} + \frac{3\mu_0 I}{16R}\right)(-\hat{k})$
 4. $\left(\frac{-\mu_0 I}{4\pi R}\right)(\hat{j}) + \left(\frac{\mu_0 I}{4R} + \frac{\mu_0 I}{2\pi R}\right)(-\hat{k})$ and $\frac{\mu_0 I}{4\pi R}(\hat{j}) + \frac{\mu_0 I}{8\pi R}(-\hat{k}) + \left(\frac{\mu_0 I}{8R} + \frac{3\mu_0 I}{8R}\right)(-\hat{k})$
2. There exist a uniform magnetic field of 1T and a uniform electric field of 1V/m – both along $(-\hat{i} + \hat{j})$. A charged particle of mass 1 kg and of charge 1C is having a velocity of $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ m/s and is at origin at $t = 0$. Then, the coordinates of the particle at time ' π ' seconds will be (Take $\pi^2 = 10$)
1. $(0, 5, 2)\text{m}$
 2. $\left(0, \frac{5}{\sqrt{2}}, -2\right)\text{m}$
 3. $\left(-\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}}, 2\right)\text{m}$
 4. $(-5, 5, -2)\text{m}$

3. A conducting rod of mass m and length l is placed on rough conducting rails having coefficient of friction $\mu = 0.25$ between the rails and the rod. Vertical supports are non-conducting. The generator G supplies current I_0 as shown in figure and uniform magnetic field exist as $\vec{B} = \frac{mg}{\ell I_0}[-\hat{i} - \hat{k}]$. If the rod is released then its acceleration will be :



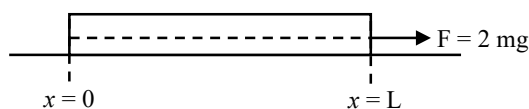
1. $\frac{g}{4}$
2. $\frac{3g}{4}$
3. $\frac{g}{3}$
4. $\frac{g}{2}$

4. The magnetic field at the centre of a current carrying square loop is of side L is B as shown in figure (i). The same current carrying loop is aligned as shown in figure (ii). If external magnetic field $B_0 \hat{i}$ exists in the region then torque experienced by current carrying loop in figure (ii) is :



1. $\frac{\pi B B_0 L^3}{4\sqrt{2} \mu_0}$ 2. $\frac{\pi B B_0 L^3}{\sqrt{2} \mu_0}$ 3. $\frac{\pi B B_0 L^3}{2\sqrt{2} \mu_0}$ 4. $\frac{B B_0 L^3}{\mu_0}$
5. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is $4/3$. A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant, When the ball is 12.8 m above the water surface, the fish sees the speed of ball as [Take $g = 10 \text{ m/s}^2$]
1. 9 m/s 2. 12 m/s 3. 16 m/s 4. 21.33 m/s
6. In the situation shown in the adjacent figure, what is the maximum elongation produced in the spring if initially the spring is relaxed (consider all contact surfaces to be smooth)?
-
1. $\frac{F}{K}$ 2. $\frac{F}{2K}$ 3. $\frac{2F}{K}$ 4. $\frac{4F}{K}$
7. A particle of mass 2g and charge $1\mu\text{C}$ is projected from origin with velocity $((2\hat{i} + 2\hat{j}) \text{ m/s})$. If potential in the region is given by $V = -(x^2 + 4x)$ kilovolts. What is trajectory of particle?
1. $y = 2 \ln(x + 2)$ 2. $y = \ln\left(\frac{x + 2}{2}\right)$
 3. $y = 2 \ln\left(\frac{x + 2}{2}\right)$ 4. $y = \ln(x + 2)$
8. Which of following is incorrect?
1. The minimum sum of the distances of a real object and a real image from a concave mirror of radius of curvature 'R' is $2R$.
 2. For a concave mirror of radius of curvature 'R', the distance between the two positions of the object for which the image size is double of the object size is $2R$.
 3. The minimum distance of the real image of a real object, formed by a concave mirror of radius of curvature 'R' from the mirror is $R/2$.
 4. The largest distance of the image of real object from a convex mirror of radius of curvature 'R' can be 'R' from the mirror.

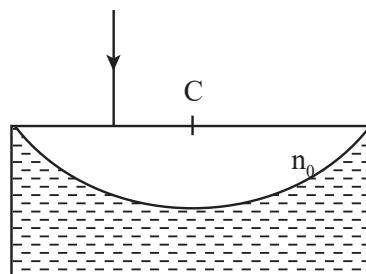
9. An uniform rope of total mass m is placed on a rough horizontal surface and is being pulled by a constant horizontal force $F = 2mg$. The coefficient of friction



between the rope and horizontal surface depends on x -coordinate as $\mu = 0.5 \frac{x}{L}$. If the tension at mid-point of the rope is given by

1. $\frac{15mg}{16}$ 2. $\frac{5mg}{8}$ 3. $\frac{5mg}{16}$ 4. $\frac{5mg}{4}$

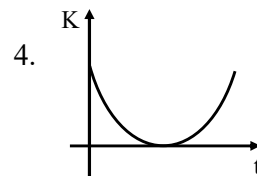
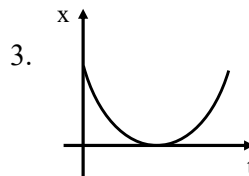
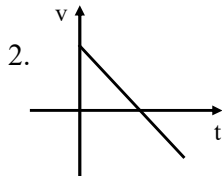
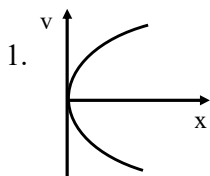
10. There is a hemisphere of radius R and refractive index n_0 . Its curved surface is followed by another medium. A ray is normally incident on the flat surface of the hemisphere at a distance $\frac{\sqrt{3}R}{2}$ from the centre C .



If this ray has to suffer total internal reflection at the curved interface, the maximum value of the refractive index of the following medium should be

1. $\sqrt{3} n_0$ 2. $\frac{\sqrt{3}}{2} n_0$ 3. $\sqrt{2} n_0$ 4. $\frac{n_0}{2}$

11. A particle moves along x -axis then among the given four graphs. One graph cannot be related to other three graphs. Choose the graph: ($v \rightarrow$ velocity, $K \rightarrow$ kinetic energy, $x \rightarrow$ position)



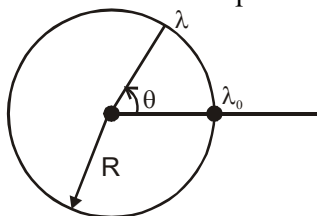
12. A ball is projected near the surface with speed u at an angle θ with the horizontal surface of an imaginary planet whose acceleration due to gravity on its surface is $g_p = 2g$ and wind blows with an acceleration of $2g$ along horizontal in the direction of throw. Find the range of the projectile. ($g =$ acceleration due to gravity on the earth's surface)

1. $\frac{2u^2}{g} \sin \theta (\sin \theta + \cos \theta)$ 2. $\frac{u^2}{2g} \cos \theta (\sin \theta + \cos \theta)$
3. $\frac{u^2}{g} \cos \theta (\sin \theta + \cos \theta)$ 4. $\frac{u^2}{g} \sin \theta (\sin \theta + \cos \theta)$

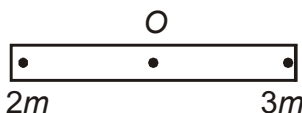
13. A toy car revolves in a circular path of radius 1m on a horizontal rough plane. Its speed varies as $V = 2t^2$. It starts sliding at $t = 1\text{s}$. Find the value of coefficient of friction between ground and the wheels of the car. Use $g = 10 \text{ m/s}^2$.

1. $\frac{2\sqrt{2}}{5}$ 2. $\frac{2}{5}$ 3. $\frac{4}{5}$ 4. $\frac{3\sqrt{2}}{5}$

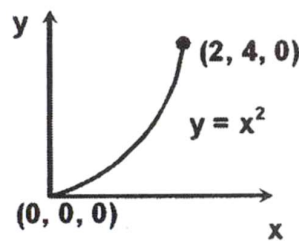
14. A thin non-conducting ring of radius R has a linear charge density $\lambda = \lambda_0 \cos \theta$, where θ is measured as shown. The total electric dipole moment of the charge distribution is:



1. $R^2 \lambda_0$ 2. $2\pi R^2 \lambda_0$ 3. $\pi R^2 \lambda_0 / 2$ 4. $\pi R^2 \lambda_0$
15. Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are same. The two bodies emit total radiant power at the same rate. The wavelength λ_B corresponding to maximum spectral radiance from B is shifted from the wavelength corresponding to maximum spectral radiance in the radiation from A by $1.0 \mu\text{m}$. If the temperature of A is 5802 K, calculate find wavelength λ_B in μm .
1. 0.5 2. 1 3. 1.5 4. 2
16. A rigid container, containing 3 moles of an ideal diatomic gas is heated in such a way that the temperature of the gas increases to 400 K from 300 K. Also, one mole of the diatomic gas breaks into individual free atoms. Find the net heat transferred to the gas.
1. 950 R 2. 450 R 3. 800 R 4. 400 R
17. A light ray incident along vector $2\hat{i} + 4\hat{j} + \sqrt{5}\hat{k}$ strikes on the x-z plane from medium I of refractive index $\sqrt{3}$ and enters into medium II of refractive index is μ_2 . The value of μ_2 for which the value of angle of refraction becomes 90° , is
1. $\frac{4\sqrt{3}}{5}$ 2. $\frac{3\sqrt{3}}{5}$ 3. $\frac{2\sqrt{3}}{5}$ 4. $\frac{\sqrt{3}}{5}$
18. There is a uniform bar of mass m and length L hinged at centre O so that it can turn in a vertical plane about a horizontal axis. Two point masses $2m$ and $3m$ are rigidly attached to its ends and the arrangement is released from rest, in horizontal position as shown. The initial angular acceleration of the bar is

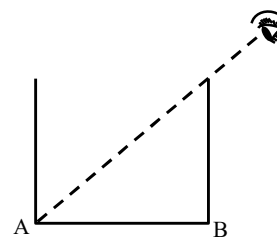


1. $\frac{3g}{8L}$ 2. $\frac{2g}{5L}$ 3. $\frac{2g}{3L}$ 4. $\frac{5g}{8L}$
19. By applying a force $\vec{F} = ((3xy - 5z)\hat{j} + 4z\hat{k})$ N a particle is moved along the path $y = x^2$ from point $(0, 0, 0)$ m to the point $(2, 4, 0)$ m. The work done by the F on the particle is
1. $\frac{280}{5}$ J 2. $\frac{140}{5}$ J 3. $\frac{232}{5}$ J 4. $\frac{192}{5}$ J



20. A cylindrical container has diameter and depth both equal to h . A coin is placed at the centre of the bottom of the container. The eye is so adjusted that the corner A is just visible. Now the container is completely filled with a liquid, the coin becomes just visible. The refractive index of the liquid is :

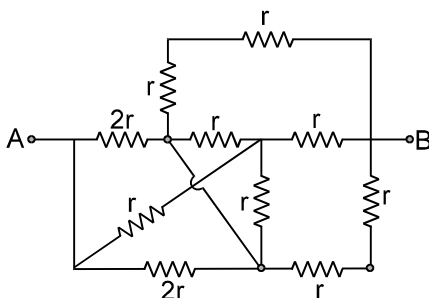
1. $\sqrt{2}$
2. $\sqrt{3}$
3. $\sqrt{\frac{5}{2}}$
4. $\sqrt{\frac{7}{2}}$



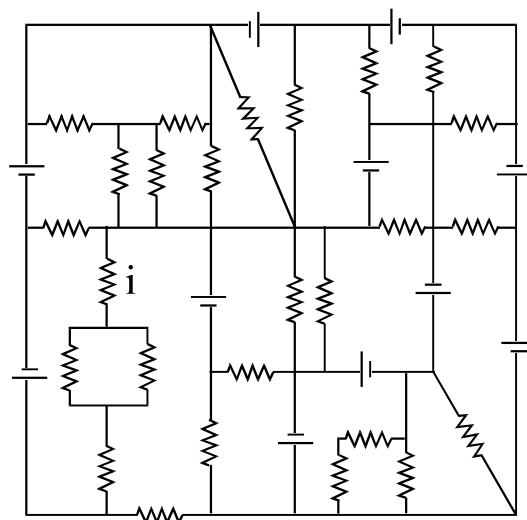
SECTION – II (Total Marks : 20)
(Integer Answer Type)

This section contains **5 questions**. This answer to each of the questions is **integer Type** ranging from 0 to 999. The bubble corresponding to the correct answer is to be darkened in the ORS.

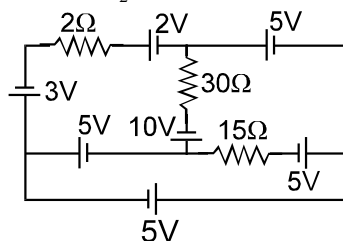
21. If $r = 2\Omega$, the equivalent resistance between A and B in the arrangement of resistances as shown, is Ω



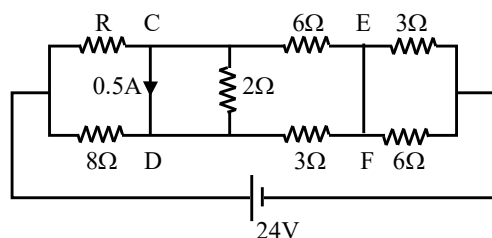
22. What are the current i (in A) in figure, where all resistances are 4.0Ω and all batteries have an emf of 10 V ?



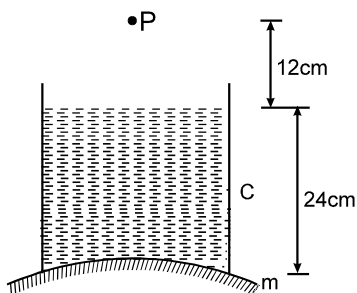
23. In the circuit shown, current through the resistance 2Ω is i_1 and current through the resistance 30Ω is i_2 . Find the ratio $\frac{i_1}{i_2}$.



24. In the given circuit 'R' is equal to Ω



25. An insect at point 'P' sees its two images in the water mirror system as shown in the figure. One image is formed due to direct reflection from water surface and the other image is formed due to refraction, reflection & again refraction by water mirror system in order. Mirror has focal length 60 cm. If the separation between the two images is d cm then find d . ($\mu_w = 4/3$).



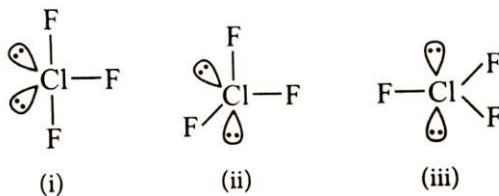
PART – B (CHEMISTRY)
SECTION – I (Total Marks : 80)
(Single Correct Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONLY ONE is correct**

26. Given below is the table showing shapes of some molecules having lone pairs of electrons. Fill up the blanks left in it and select the correct option.

Molecule type	bp	lp	Shape	Example
AB_2E_2	2	<u>P</u>	Bent	H_2O
AB_3E_2	3	<u>Q</u>	<u>Q</u>	ClF_3
AB_5E	5	<u>R</u>	<u>S</u>	BrF_5
AB_4E_2	4	2	<u>T</u>	<u>U</u>

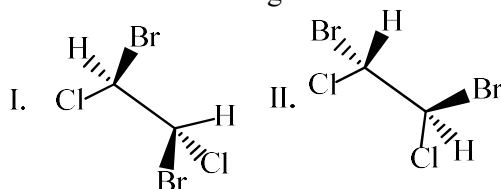
- | | P | Q | R | S | T | U |
|----|---|------------------|---|------------------|------------------|----------|
| 1. | 2 | Square pyramidal | 2 | T-shaped | Square planar | H_2O_2 |
| 2. | 4 | T-shaped | 5 | Square planar | Square pyramidal | SO_3 |
| 3. | 2 | T-shaped | 1 | Square pyramidal | Square planar | XeF_4 |
| 4. | 3 | Square planar | 2 | T-shaped | Square pyramidal | $BrCl_3$ |
27. The most stable shape of ClF_3 is shown by



1. only (i) 2. (i) and (iii) 3. only (ii) 4. Only (iii)
28. The increasing order of energies of various molecular orbitals of N_2 is given below:
 $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$
 The above sequence is not true for the molecule
1. C_2 2. B_2 3. O_2 4. Be_2
29. Among the following, which are ambidentate ligands.
- (i) SCN^- (ii) NO_3^- (iii) NO_2^- (iv) $C_2O_4^{2-}$
1. (i) and (iii) 2. (i) and (iv) 3. (ii) and (iii) 4. (ii) and (iv)
30. The charges x and y on the following ions are
- (i) $[Co(NH_3)_2Cl_4]^x$ (ii) $[Fe(CN)_6]^y$
 (Oxidation state of Co is +3 and Fe is +2 in their respective complexes.)
1. $x = +1, y = -1$ 2. $x = -1, y = +3$
 3. $x = -1, y = -4$ 4. $x = -2, y = -3$

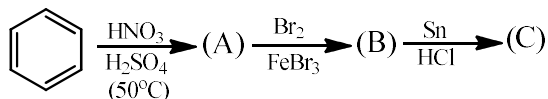
31. $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ show different colours in dilute solution because
1. CN^- is a strong field ligand and H_2O is a weak field ligand hence magnitude of CFSE is different.
 2. both CN^- and H_2O absorb same wavelength of energy.
 3. complexes of weak field ligands are generally colourless
 4. the sizes of CN^- and H_2O are different hence their colours are also different.

32. Consider the following



Which of the following statement is incorrect for above compound

1. I is chiral and II is achiral
 2. I and II are Diastereomers
 3. I is achiral and II is chiral
 4. I is meso and II is optically active
33. Which pair represents chain isomers?
1. Ethanol and Dimethyl ether
 2. n-Butane and isobutane
 3. Propanal and Propanone
 4. But-1-ene and But-2-ene
34. Which of the following compound cannot show geometrical isomerism?
1. 3-ethylhex-2-ene
 2. 2-methylpent-2-ene
 3. 1-chloropent-2-ene
 4. 1,2-dichlorocyclopentane
35. Identify (A), (B) and (C) in the following sequence of reactions.



1. A $\rightarrow$ Nitrobenzene, B $\rightarrow$ Dinitrobenzene, C $\rightarrow$ p-Bromoaniline
 2. A $\rightarrow$ $\text{C}_6\text{H}_5\text{SO}_3\text{H}$, B $\rightarrow$ m-Benzenesulphonic acid, C $\rightarrow$ m-Benzenesulphonate
 3. A $\rightarrow$ $\text{C}_6\text{H}_5\text{NO}_2$, B $\rightarrow$ m-Bromonitrobenzene, C $\rightarrow$ m-Bromoaniline
 4. A $\rightarrow$ p-Nitrobenzene, B $\rightarrow$ m-Trinitorbenzene, C $\rightarrow$ m-Bromoaniline
36. The reaction in terms of intermediates and type of reaction is given below. Mark the correct option(s).

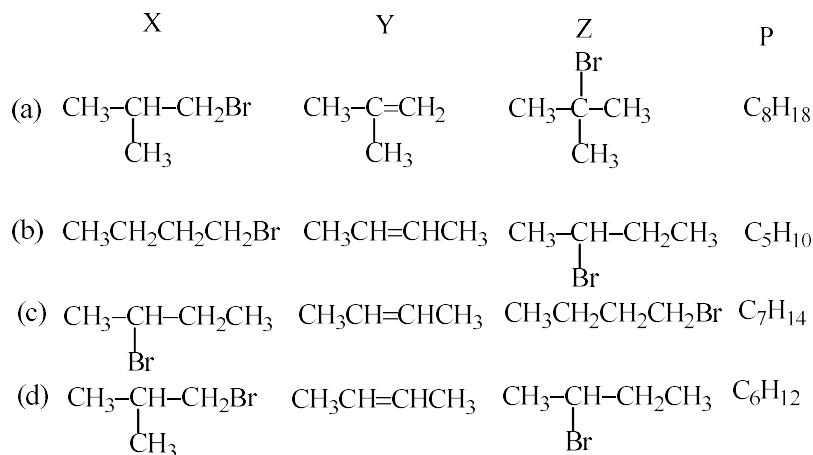
- I. $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2 + \text{HBr} \rightarrow \text{Carbocation intermediate}$
- II. $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2 + \text{HBr} \xrightarrow{\text{Peroxide}} \text{Free radical intermediate}$
- III. $>\text{C}=\text{C}< + \text{X}_2 \rightarrow \text{electrophilic substitution}$
- IV. $>\text{C}=\text{C}< + \text{O}_3 \rightarrow \text{Ozonide}$

1. III only
2. II and IV only
3. I, II and IV only
4. I, II and III only

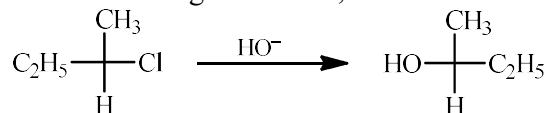
37. Match the reactions given in Column I with the type of reaction mentioned in Column II and mark the appropriate choice.

	Column I		Column II
A.	$\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{Br} \quad \text{CH}_3 \end{array} \xrightarrow{\text{C}_2\text{H}_5\text{OH}} \begin{array}{c} \text{OC}_2\text{H}_5 \\ \\ \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	i.	β -elimination
B.	$\text{CH}_3\text{CH}_2\text{Br} \xrightarrow{\text{aq. KOH}} \text{CH}_3\text{CH}_2\text{OH}$	ii.	S_N^1 nucleophilic substitution
C.	$\text{CH}_3\text{CH}=\text{CH}_2 + \text{HBr} \xrightarrow{\text{Peroxide}} \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	iii.	S_N^2 nucleophilic substitution
D.	$\text{CH}_3-\text{CH}_2\text{Br} + \text{alc. KOH} \rightarrow \text{CH}_2=\text{CH}_2$	iv.	Kharasch effect

1. A-iv, B-i, C-ii, D-iii 2. A-ii, B-iii, C-iv, D-i
 3. A-i, B-ii, C-iv, D-iii 4. A-iii, B-i, C-ii, D-iv
38. Primary alkyl halide, $\text{C}_4\text{H}_9\text{Br}$ (X) reacts with alc. KOH to give compound (Y). (Y) reacts with HBr to give compound (Z) which is an isomer of (X). When (X) reacts with Na metal it gives compounds (P). (X), (Y), (Z) and (P) are

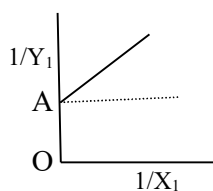


1. (a) 2. (b) 3. (c) 4. (d)
39. In the reaction given below,



Which of the following statements is correct?

- The reaction proceeds via S_N^2 mechanism hence inversion of configuration takes place.
- The reaction proceeds via S_N^1 mechanism hence inversion of configuration takes place.
- The reaction proceeds via S_N^2 mechanism hence there is no change in the configuration.
- The reaction proceeds via S_N^1 mechanism hence there is no change in the configuration.

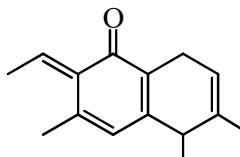
40. How many moles of NaOH can be added to 1.00 L of a solution of 0.10 M in NH_3 and 0.10 M in NH_4Cl without changing the pOH by more than 1.00 unit (assume no change in volume)? $K_b(\text{NH}_3)=1.8\times 10^{-5}$, $\text{p}K_b=4.74$
 1. 0.182 2. 0.082 3. 0.200 4. 0.282
41. Given, $\text{H}_2\text{S}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HS}^-(\text{aq})$; $K = 9.1\times 10^{-8}$ and for $\text{FeS}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{Fe}^{2+}(\text{aq}) + \text{HS}^-(\text{aq}) + \text{OH}^-(\text{aq})$; $K = 5 \times 10^{-19}$
 Thus, molar solubility of $\text{FeS}(\text{s})$ in a buffer solution of $\text{pH} = 3.0$ in which $[\text{H}_2\text{S}]$ is fixed at 0.1 M, is
 1. $5.5 \times 10^{-3}\text{M}$ 2. $1.2 \times 10^{-2}\text{M}$ 3. $9.1 \times 10^{-8}\text{M}$ 4. $7.9 \times 10^{-7}\text{M}$
42. 1 molal solution of each of AB and CD are in same solvent X, AB is ionised in X.
 $\text{AB} \rightleftharpoons \text{A}^+ + \text{B}^-$ (α =degree of ionisation)
 CD is dimerised in X.
 $\text{CD} \rightleftharpoons \frac{1}{2}(\text{CD})_2$ (β = degree of dimensation)
 If, $\Delta T_b(\text{AB})=2 \Delta T_b(\text{CD})$, then $(\alpha+\beta)$ is
 1. 0.2 2. 0.4 3. 0.8 4. 1.0
43. On addition of a solute, the vapour pressure of a liquid reduces to $\left(\frac{9}{10}\right)^{\text{th}}$ of the original value. If 2g of the solute (molar mass 100g mol^{-1}) is added to 100 g of the liquid to attain the reduction, then molar mass of the liquid is
 1. 200 g mol^{-1} 2. 555 g mol^{-1}
 3. 500 g mol^{-1} 4. 400 g mol^{-1}
44. A mixture of two volatile components (1) and (2) has mole fraction of component (1) as X_1 in liquid phase and as Y_1 in vapour phase.
 A graph between $\left(\frac{1}{Y_1}\right)$ along (y-axis) and $\left(\frac{1}{X_1}\right)$ along x-axis is 
- linear (p_1° and p_2° are vapour pressure of components (1) and (2) in pure state)
 Thus, slope and intercept OA are respectively
- | | Slope | Intercept OA | | Slope | Intercept OA |
|----|---|---|----|---|---|
| 1. | $\frac{(p_1^\circ - p_2^\circ)}{p_2^\circ}$ | $\frac{p_2^\circ}{p_1^\circ}$ | 2. | $\frac{(p_1^\circ - p_2^\circ)}{p_1^\circ}$ | $\frac{p_2^\circ}{p_1^\circ}$ |
| 3. | $\frac{p_1^\circ}{p_2^\circ}$ | $\frac{(p_1^\circ - p_2^\circ)}{p_1^\circ}$ | 4. | $\left(\frac{p_2^\circ}{p_1^\circ}\right)$ | $\frac{(p_1^\circ - p_2^\circ)}{p_1^\circ}$ |
45. Two radioactive substances A and B have disintegration constants as 8λ and λ respectively. Initially, both the substances have equal number of molecules. After time t,
 $N_B/N_A = e$
 Thus, time t is
 1. $\frac{1}{9\lambda}$ 2. $\frac{1}{\lambda}$ 3. $\frac{1}{7\lambda}$ 4. $\frac{1}{8\lambda}$

SECTION – II (Total Marks : 20)
(Integer Answer Type)

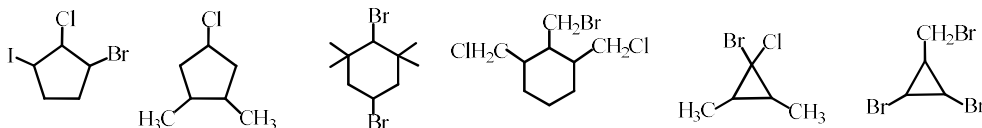
This section contains **5 questions**. This answer to each of the questions is **integer Type** ranging from 0 to 999. The bubble corresponding to the correct answer is to be darkened in the ORS.

46. The terminal and bridged CO ligands in the compound $[\text{Co}_2(\text{CO})_8]$ are n and m respectively. Find the value of $(n \times m)$

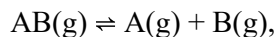
47. How many Deuterium can be incorporated in the given molecule assuming excess and prolonged treatment with $\text{OD}^- / \text{D}_2\text{O}$



48. In how many compounds one of the stereoisomers will be meso.

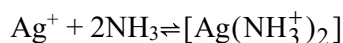


49. For the equilibrium,



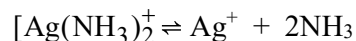
the equilibrium pressure at which one third of AB is dissociated, is n times of equilibrium constant. What is value of n.

50. For the complex,



$$\left(\frac{dx}{dt} \right) = 2 \times 10^7 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1} [\text{Ag}^+][\text{NH}_3]^2 - 1 \times 10^{-2} \text{ s}^{-1} [\text{Ag}(\text{NH}_3)_2]^+$$

Thus, for the reaction



equilibrium constant is $X \times 10^{-10}$. The value of X will be

PART – C (MATHEMATICS)
SECTION – I (Total Marks: 80)
(Single Correct Answer Type)

This section contains 20 multiple choice questions. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct

51. If $\vec{r} = \alpha \vec{b} \times \vec{c} + \beta \vec{c} \times \vec{a} + \gamma \vec{a} \times \vec{b}$ and $[\vec{a} \ \vec{b} \ \vec{c}] = 2$, then $\alpha + \beta + \gamma =$
1. $\vec{r} \cdot (\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b})$
 2. $\frac{1}{2} \vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$
 3. $2\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$
 4. 4
52. A number x is chosen at random from the set $\{1, 2, 3, 4, \dots, 100\}$. Then the probability of selecting x which satisfy the inequality $\frac{(50-x)(10-x)}{(30-x)} \leq 0$ is
1. 0.20
 2. 0.51
 3. 0.70
 4. 0.71
53. Let ABC be a triangle such that $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{AC} = \vec{b}$, $\overrightarrow{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 12$. Consider the statements:
- (S1) : $|(\vec{a} \times \vec{b}) + (\vec{c} \times \vec{b})| - |\vec{c}| = 6(2\sqrt{2} - 1)$ (S2) : $\angle ABC = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$. Then
1. both (S1) and (S2) are true
 2. Only (S1) is true
 3. only (S2) is true
 4. Both (S1) and (S2) are false
54. Let the points on the plane P be equidistant from the points $(-4, 2, 1)$ and $(2, -2, 3)$. If $\langle l, m, n \rangle$ are direction ratios of the line of intersection plane P and the plane $2x + y + 3z = 1$ then $\frac{l-m+n}{2l+m-3n} =$
1. $-1/3$
 2. $1/3$
 3. $1/6$
 4. $-1/6$
55. Consider a ΔABC where $A(1, 3, 2)$, $B(-2, 8, 0)$ and $C(3, 6, 7)$. If the angle bisector of $\angle BAC$ meets the line BC at D. then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is.
1. $\frac{37}{2\sqrt{38}}$
 2. $\sqrt{9}$
 3. $\frac{39}{2\sqrt{38}}$
 4. $\frac{\sqrt{38}}{2}$
56. If the system of equations
- $$-kx + 3y - 14z = 25, \quad -15x + 4y - kz = 3 \text{ and } -4x + y + 3z = 4$$
- is consistent for $k \in \{2n + 1 : n = -6, -5, -4, 3, 4, 5\}$ then the sum of divisors of N , where $N = \sum(k^2 - k + 1)$ is
1. 648
 2. 720
 3. 896
 4. 696

57. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, ($\alpha \in \mathbf{R}$) such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then a value of α is:

1. $\frac{\pi}{64}$ 2. 0 3. $\frac{\pi}{32}$ 4. $\frac{3\pi}{64}$

58. Let k be a positive real number and let $A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$ and

$$B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}. \text{ If } \det(\text{adj}A) + \det(\text{adj}B) = 20^6, \text{ then } k =$$

1. $9/2$ 2. $19/2$ 3. $21/2$ 4. $29/2$

59. The probability that student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has a 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Then the correct relation is

1. $p + m + c = \frac{19}{20}$ 2. $p + m + c = \frac{27}{20}$
3. $pmc = \frac{1}{4}$ 4. $mp + mc + pc - 2mpc = 0.4$

60. Let D be the domain of the function $f(x) = \sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $g: D \rightarrow \mathbf{R}$ defined by $g(x) = x - [x]$, ($[x]$ is the greatest integer function), is (α, β) , then $\alpha^2 + \frac{5}{\beta}$ is equal to

1. 135 2. 46 3. 45 4. 136

61. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a function defined by $f(x) = (2+3a)x^2 + \left(\frac{a+2}{a-1}\right)x + b$, $a \neq 1$.

If $f(x+y) = f(x) + f(y) + 1 - \frac{2}{7}xy$, then the value of $28 \sum_{i=1}^5 |f(i)|$ is

1. 735 2. 675 3. 545 4. 715

62. Let $f: \mathbf{R} \rightarrow [2, \infty)$ be a function defined by $f(x) = x^2 - 12ax + 15 - 2a + 36a^2$. If $f(x)$ is surjective on $\mathbf{R}$ then value of a is equal to

1. $\frac{9}{2}$ 2. $\frac{11}{2}$ 3. $\frac{13}{2}$ 4. $\frac{15}{2}$

63. Let S be the set of all $a \in \mathbf{N}$ such that the area of the triangle formed by the tangent at the point $P(b, c)$, $b, c \in \mathbf{N}$ on the parabola $y^2 = 2ax$ and the lines $x = b$, $y = 0$ is 16 units², then $\sum_{a \in S} a$ is equal to

1. 138 2. 141 3. 143 4. 146

64. Let the shortest distance from $(a, 0)$, $a > 0$, to the parabola $y^2 = 4x$ be 4. Then the equation of the circle passing through the point $(a, 0)$ and the focus of the parabola, and having its centre on the axis of the parabola is
1. $x^2 + y^2 - 4x + 3 = 0$
 2. $x^2 + y^2 - 8x + 7 = 0$
 3. $x^2 + y^2 - 10x + 9 = 0$
 4. $x^2 + y^2 - 6x + 5 = 0$
65. If the domain of the function $\log_5(18x - x^2 - 77)$ is (α, β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2 + 3x - 2}{x^2 - 3x - 4}\right)$ is (γ, δ) then $\alpha^2 + \beta^2 + \gamma^2$ is equal to
1. 195
 2. 174
 3. 186
 4. 179
66. Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = fog(10)$, $b = gof(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + l^2$ is equal to
1. 6
 2. 12
 3. 8
 4. 16
67. Let the acute angle bisector of the two planes $x - 2y - 2z + 1 = 0$ and $2x - 3y - 6z + 1 = 0$ be the planes P. Then which of the following points lies on P?
1. $\left(3, 1, -\frac{1}{2}\right)$
 2. $(4, 0, -2)$
 3. $(0, 2, -4)$
 4. $\left(-2, 0, -\frac{1}{2}\right)$
68. Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1 , then the sum of the roots of $f(x - 2) = 0$ is equal to :
1. $\frac{11}{3}$
 2. $\frac{7}{3}$
 3. $\frac{23}{3}$
 4. $\frac{14}{3}$
69. Consider two vectors $\vec{a}$ and $\vec{b}$. If $3\vec{a} = 2\vec{i} - \vec{j} - 2\vec{k}$ and $5\vec{b} = -4\vec{j} + 3\vec{k}$ Then $|(\vec{a} + \vec{b}) + 2(\vec{a} \times \vec{b})| =$
1. $\frac{7\sqrt{26}}{15}$
 2. $\frac{11\sqrt{6}}{15}$
 3. $\frac{7\sqrt{23}}{15}$
 4. $\frac{11\sqrt{3}}{15}$
70. Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α . If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to:
1. 37
 2. 58
 3. 68
 4. 92

SECTION – II (Total Marks : 20)
(Integer Answer Type)

This section contains **5 questions**. This answer to each of the questions is **integer Type** ranging from 0 to 999. The bubble corresponding to the correct answer is to be darkened in the ORS.

71. the point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $A(-3, -6, 1)$ and $B(2, 4, -3)$ divides the line segment AB internally in the ratio $K : 1$. If a, b, c ($|a|, |b|, |c|$ are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $2|a + b + c| - 13 =$
72. Let the system of linear equations $x + y + \alpha z = 2$, $3x + y + z = 4$ and $x + 2z = 1$ have a unique solution (p, q, r) . If (α, p) , (q, α) and $(-2p, -q)$ are collinear points, then the sum of absolute values of possible values of α is:
73. If the radius of a circle touching the pair of lines $7x^2 - 18xy + 7y^2 = 0$ and the circle $x^2 + y^2 - 8x - 8y = 0$, and contained in the given circle is equal to k , then $\frac{k^2}{2}$ is equal to
74. Let A be the event of selecting two real numbers x, y from the set $[0, 60]$ satisfying $|x - y| \leq k$. If $P(A) = \frac{11}{36}$ then $99k$ is equal to
75. If $f(x + y) = f(xy) \forall x, y \in \mathbf{R}$, and $f(2003) = 2003$, then $\frac{f(-2003) - 2000}{3}$ equals